

dependencies to be used by the project. Here we list 'grunt' and its version, and in this example we will use a Grunt plugin to 'uglify' our JavaScript; so we list that too along with its version.

Next, it is time to install all the required modules. To do this, simply run the following command:

```
npm install
```

Please ensure that you are at the root directory of the project. Once the installation is complete, you can see a new folder called 'node_modules' being created in your project, as shown in Figure 2. As you can see inside this folder, all the node modules that we had specified in the *package.json* file have been installed. This is how we actually install Grunt for our project.

Now that we have installed Grunt and all required Grunt plugins for our project, let's start configuring it in order to be able to use it. To do this, we need to create a *Gruntfile.js* in our project as already discussed above. *Gruntfile.js* is a JavaScript file, using which we can configure Grunt. This file has to be at the root directory of the project alongside the *package.json* file. And it should be named as it is.

```
module.exports = function(grunt) {
  /** Project configuration
  grunt.initConfig({
    pkg: grunt.file.readJSON('package.json'),
    uglify: {
      options: {
        banner: '/*! Project: <%= pkg.name %> | Version #: <%=
pkg.version %> | Created: <%= grunt.template.today("yyyy-mm-
dd") %> */\n'
      },
      build: {
        src: 'src/<%= pkg.name %>ScriptFile.js',
        dest: 'build/<%= pkg.name %>ScriptFile.min.js'
      }
    }
  });

  /** Load the grunt-contrib-uglify plugin, using which we
will uglify the JavaScript files
  grunt.loadNpmTasks('grunt-contrib-uglify');

  /** Register the default task.
  grunt.registerTask('default', ['uglify']);
};
```

The entire code for the *Gruntfile.js* is inside one function called *module.exports*. Inside this function, we make a call to the *grunt.initConfig* method. Through this method, we can instruct Grunt about what we want it to do. First, we tell it to read project configurations from the *package.json*. This way,

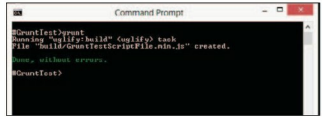


Figure 3: Running Grunt

we can use the metadata stored in the *package.json* files and refer to it in the *Gruntfile.js*. So to access the project name defined in the *package.json*, we can simply write *<%= pkg.name %>*. Similarly, we can access all the keys that we had defined in the *package.json* file.

Next, we set configurations for the *uglify* task. We can put up a banner that will act as a comment on the top to the *uglified* JavaScript. Then we specify the *uglify* build options, by telling the plugin the location of the files to be *uglified*, and the destination for placing the *uglified* files. As you can see, here we have made use of the metadata and instead of writing *GruntTestScriptFile.js*, we wrote *<%= pkg.name %>ScriptFile.js*, though there is no mandate for following such a method. In real-time scenarios, the name will have to be typed explicitly instead of using the metadata.

Next, we load the *uglify* plugin and register it as a default task. The default task can be listed in an array, so this array can contain 'n' values based on all the plugins that one would want to use.

Now, everything is in place, and we can go ahead and use Grunt to automate our task of *uglify*ation. To run Grunt, simply type the following in the command prompt:

```
grunt
```

If everything was done properly, this will execute and show a result like the one shown in Figure 3. Now, if you check your folder structure, there will be an *uglified* version of the 'GruntTestScriptFile.js' created in the 'build' folder. So every time you make any changes to this file and need to run the task of *uglify*ation, just type *grunt* at the command prompt and you are good to go. When working with many tasks related to organising, testing and maintaining, this simple command *grunt* will simplify the work of the programmer.

Grunt has a lot of depth to it, and has many plugins that perform various tasks. One can read more about Grunt at <http://gruntjs.com/>. 

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